

Name: _____

Date: _____

Standards: MA.5.FR.1.1; MA.5.FR.2.1-2.4; MA.5.AR.1.2-1.3

Show your work in the box for each problem.

Write your final answer on the answer line (or in the blanks shown).

Use the words and strategies from Unit 05: multiplication of fractions.

1 Fraction × Whole Number

Find each product. Simplify.

$$\frac{3}{5} \times 10 = \underline{\hspace{2cm}} \quad | \quad \frac{7}{8} \times 16 = \underline{\hspace{2cm}} \quad | \quad \frac{4}{9} \times 27 = \underline{\hspace{2cm}}$$

Answers _____

2 Fraction × Fraction

Multiply. Write in simplest form.

$$\frac{3}{4} \times \frac{2}{5} = \underline{\hspace{2cm}} \quad | \quad \frac{5}{6} \times \frac{3}{7} = \underline{\hspace{2cm}} \quad | \quad \frac{8}{9} \times \frac{3}{4} = \underline{\hspace{2cm}}$$

Answers _____

3 Mixed Number × Fraction

Multiply. Simplify.

$$2 \frac{1}{3} \times \frac{3}{4} = \underline{\hspace{2cm}} \quad | \quad 3 \frac{1}{2} \times \frac{2}{5} = \underline{\hspace{2cm}}$$

Answers _____

7 Find the Mistake

Kai multiplies $2\frac{1}{2} \times 1\frac{3}{4}$. He computes: $2 \times 1 = 2$, and $\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$, so the answer is $2\frac{3}{8}$.

Is Kai correct? Convert both to improper fractions and multiply correctly. Explain why Kai's method fails.

Correct Answer and Explanation _____

8 Garden Area

A rectangular garden measures $4\frac{2}{3}$ meters by $3\frac{3}{5}$ meters.

Find the area of the garden in square meters. Show your complete work.

Answer _____

9 Three-Factor Multiplication

Find: $\frac{3}{4} \times \frac{2}{3} \times \frac{5}{6}$.

Show your work in two ways: (1) left to right, (2) cross-simplify first. Do you get the same answer?

Answers and Comparison _____

10 Predict and Prove

Without calculating, rank these from least to greatest: $\frac{4}{5} \times \frac{3}{5}$, $\frac{4}{5} \times 1$, $\frac{4}{5} \times \frac{6}{5}$.

State your ranking with reasoning, then verify each product. Was your prediction correct?

Ranking, Verification, and Explanation _____